

NARENDRAN PALANIVEL

Data Analyst

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PROFESSIONAL SUMMARY

Accumulated as a Data Analyst with **around 3years** of hands-on experience delivering end-to-end data solutions across **pharmaceutical, fintech, and advertising sectors**. **Expert in building scalable ETL** pipelines using **Informatica, AWS Glue, Azure Data Factory, and Airflow** to **streamline data integration** from disparate sources. Adept at crafting advanced visualizations and interactive dashboards in Power BI and Tableau to drive strategic insights for business units. Skilled in **Python, SQL, and R** for advanced analytics, predictive modeling, and automation—leveraging libraries like **Scikit-learn, Pandas, and TensorFlow** to implement regression, classification, and anomaly detection models. Proven track record in executing data validation, fraud detection, and compliance automation to improve system reliability and decision accuracy. Led cross-functional initiatives using **Agile methodologies**, integrating cloud platforms (**AWS, Azure, GCP**) and **DevOps** tools to support continuous data delivery. Project experience includes credit risk scoring, drone-based surveillance using **YOLOv9**, and IoT-enabled smart home monitoring, demonstrating a strong blend of technical acumen, domain adaptability, and innovation in real-world deployments.

EDUCATION

The University of Texas

Master of Science in Data Science

Arlington, TX, USA

August 2023 - May 2025

Amrita Vishwa Vidyapeetham

Bachelor's of Technology in Electricals and Electronics Engineering

India

Jun 2017 - May 2021

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, HTML, JavaScript, R, Shell Scripting

Python Libraries & Frameworks: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Matplotlib, Seaborn, Plotly, NLTK, Statsmodels

Databases: MySQL, PostgreSQL, Oracle, MongoDB, SQLite, Snowflake

Data Visualization Tools: Power BI, Tableau, Microsoft Excel (Advanced Charts, Pivot Tables), Plotly

Analytics & Statistical Skills: Exploratory Data Analysis (EDA), Data Mining, Data Cleaning, Feature Engineering, Hypothesis Testing, A/B Testing, Time Series Analysis, Text Mining, Predictive Modeling

ETL & Data Engineering: Apache Airflow, AWS Glue, Informatica, DBT, Data Warehousing (Star/Snowflake Schema), SQL-based ETL pipelines

Cloud Platforms & DevOps Tools: AWS (S3, Lambda, RDS, Redshift), Google Cloud (BigQuery), Azure, Jenkins, Docker, Git, GitHub, Terraform

APIs & Scripting: REST API Integration, Web Scraping, JSON/XML Parsing, Automation with Python Scripts

Tools & IDEs : Google Colab, Jupyter Notebook, Visual Studio Code, PyCharm, GitHub

Operating Systems: Windows, MacOS, Linux/Unix

PROFESSIONAL EXPERIENCE

COGNIZANT

Programmer Analyst

India

Jun 2021– Aug 2023

- **Led** data analysis initiatives for Abbott Pharmaceuticals, focusing on Master Data Management (MDM), data integration, and SQL-based reporting to ensure enterprise-wide data consistency.
- **Engineered** and maintained ETL workflows across multi-source pipelines using Informatica and SQL, supporting seamless data migration and transformation.
- **Enhanced** performance of 50+ data pipelines, improving data accuracy and reducing processing time by 20 hours/month, directly supporting critical analytics operations.
- **Developed** analytical SQL models serving five business units, enabling actionable insights that streamlined internal processes and reporting workflows.
- **Strengthened** compliance through data governance strategies, resulting in a 15-point improvement in audit performance and reduced regulatory risk.
- **Built** automated ETL validation logic for 100,000+ credit card applications, achieving a 98% processing accuracy and bolstering system reliability.
- **Implemented** fraud detection mechanisms and eligibility verification rules using Python and SQL logic, decreasing manual reviews and improving throughput by 25%.
- **Redesigned** error-handling frameworks within data pipelines, leading to a 20% drop in system-related application rejections and improved data trustworthiness.
- **Collaborated** in Agile cross-functional teams to align data solutions with evolving business goals, ensuring timely and impactful deliveries.
- **Performed** continuous data cleansing and validation, ensuring high-integrity inputs for visualization tools like Tableau and Power BI.

- **Designed** and maintained SQL-based ETL pipelines across MySQL and Snowflake, processing 500K+ records to support data warehousing and improve query response time by 10 hours/week.
- **Automated** data preprocessing tasks using Python (Pandas, NumPy), minimizing manual cleaning efforts from 3 hours to under 20 minutes per dataset.
- **Crafted** interactive dashboards in Tableau and Power BI, delivering marketing insights that accelerated campaign deployment by 2 days.
- **Engaged** with product and business teams to define success metrics and KPIs, contributing to cross-team alignment and scalable reporting.
- **Explored** customer behavior using R and Seaborn for an EDA project, surfacing patterns that informed a targeted retention experiment for two customer segments.
- **Integrated** external survey data via REST APIs and automated Python scripts, improving update frequency and ensuring real-time insights for weekly operations meetings.
- **Built** a logistic regression model in Scikit-learn to predict customer churn, enabling the business to prioritize retention actions that avoided \$7,500 in user acquisition costs.
- **Streamlined** A/B testing analysis in Jupyter using hypothesis testing frameworks, reducing feature validation timelines by 3 days and boosting iteration speed.

PROJECT AND LEADERSHIP

Autonomous Drone Navigation and Surveillance System

- Designed a drone-based surveillance system using **Apache Kafka** for real-time streaming of video and sensor data from unmanned aerial devices.
- Applied **YOLOv9** for high-accuracy object detection, enabling real-time classification of humans, vehicles, and anomalies in aerial video feeds.
- Automated drone operations using **Python** and **Tello Py**, supporting autonomous takeoff, path control, and video data capture.
- Integrated **reinforcement learning algorithms** to allow drones to dynamically optimize flight paths based on environmental conditions and live data.
- Structured raw drone video and telemetry into analyzable formats, enabling **real-time anomaly detection** and enhanced **situational awareness** for security operations.
- Designed a scalable data pipeline to simulate edge processing and cloud-based analytics for live drone surveillance use cases.

Smart Home with Condition Monitoring

- Built an **IoT-enabled smart home system** with real-time monitoring of temperature, motion, and occupant health using embedded sensors.
- Developed a centralized data pipeline using **Python** and **SQL** for processing sensor outputs and generating actionable insights.
- Designed and implemented **ETL workflows** for continuous data transformation and storage, supporting automated reporting and alerts.
- Created interactive dashboards to visualize condition trends and track home safety indicators from remote interfaces.
- Applied **predictive analytics models** to detect abnormal behavior and enhance home automation responses.
- Optimized multi-sensor data ingestion and integration workflows to ensure real-time responsiveness and efficient analytics processing.
- **Published by Springer**, demonstrating innovation in smart home automation, predictive monitoring, and IoT-driven safety systems.

Credit Card System – Data Integration Project

- Designed and deployed ETL pipelines using Informatica PowerCenter to extract, transform, and load customer and transaction data from heterogeneous sources into a centralized data warehouse.
- Developed modular data flows for Customer Verification and Bank Account Validation, leveraging lookup transformations, conditional logic, and SQL-based filters to enforce business rules.
- Engineered eligibility rule checks using dynamic mappings to validate applicant history, including prior credit card ownership and account status.
- Implemented data validation and integrity checks using transformation logic (Expression, Filter, Joiner, Aggregator), optimizing workflows for scalable credit card application processing.
- Enhanced data reliability and regulatory compliance through workflow monitoring, robust error-handling, and audit trail logging mechanisms.